US Natural Disaster Temporal Analysis & Visualization
MileStone-3

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Table of Contents

1. Project Description	1
2. Dataset Used	1
3. Environment Setup	2
4. Data Exploration & Validation 4.1. Geospatial Format Verification 4.2. State Name Normalization (Mapping Dictionary) 4.3. Spatial Join & Merge Validation	3
5. Temporal Trend Analysis 5.1. Nationwide Distribution (Plotly Choropleth) 5.2. Top 10 State Comparative Analysis (Seaborn) 5.3. Geographic Incident Composition (Stacked Bar) 5.4. Interactive Hurricane Hotspot Detection (Folium)	 4 5 6 7
6. Visualization Standards	8
7. Conclusion	9

1. Project description

The **US Natural Disaster Geospatial Analysis project** is a data-driven study designed to visualize and analyze the geographical distribution of disaster declarations across the United States. While the previous phase focused on temporal evolution, this milestone transitions into **Geospatial Exploratory Data Analysis (EDA)** to identify regional patterns and risk concentrations.

Key Objectives:

- **Spatial Distribution:** Map total disaster frequencies across the 50 US states to identify national outliers.
- **Regional Specialization:** Analyze how specific disaster "signatures" (e.g., Fire vs. Hurricane) vary based on geographic location.
- **Interactive Hotspot Mapping:** Utilize advanced geospatial tools to pinpoint clusters for high-impact events like Hurricanes and Floods.
- **Data-Driven Geography:** Provide spatial justifications for why certain states experience higher declaration volumes compared to others.

2. Dataset Used

The project continues to utilize the **Federal Emergencies and Disasters (1953-Present)** dataset. While the dataset title mentions "Present," the records are comprehensive through the year **2017**. This historical archive is essential for benchmarking how environmental crises and federal responses are distributed geographically.

Dataset Characteristics for Geospatial Analysis:

- **Geographic Data**: Contains detailed state identifiers (state) which are validated and normalized to ensure a 100% match rate with physical boundary files (GeoJSON).
- **Categorical Data**: Includes incidentType labels (e.g., Fire, Flood, Hurricane), allowing for the grouping and aggregation of specific disaster categories per region.
- **Data Granularity**: Enables the extraction of specific time-based features like Year and Month to correlate seasonal patterns with geographic locations.
- **Optimized Raw DataSet**: We have taken cleaned data set from **Milestone 1**

3. Environment Setup

The analysis is implemented in **Python** using **Visual Studio Code (VS Code)**. While Milestone-2 utilized basic processing, this phase introduces an advanced geospatial toolset:

- **Pandas & NumPy**: For loading data, validating geographic columns, and performing spatial aggregations.
- **Matplotlib & Seaborn**: For generating high-fidelity static bar charts and geographic composition plots.
- **Plotly**: Employed to create dynamic, interactive nationwide choropleth maps.
- **GeoPandas & Folium**: Utilized for managing geometric shapes and rendering interactive browser-based maps for hotspot detection.
- **Jupyter Notebook**: Used within VS Code to maintain a structured, cell-based workflow for execution and exploration.

Following are the requirements we have installed for the complete project but for this specific milestone we have used only Pandas, Matplotlib and Jupyter Notebook.

```
numpy
pandas
matplotlib
seaborn
plotly
geopandas
folium
jupyter
```

4. Data Exploration & Validation

Before generating any geospatial visualizations, a comprehensive exploration phase was conducted to ensure the dataset met the structural requirements for geographic mapping

4.1. Geospatial Format Verification

The integrity of the spatial analysis depends on the presence of valid geographic identifiers.

- **Column Validation**: Verified the presence of the state and incidentType columns required for regional aggregation.
- **Format Standardization**: Used .astype(str).str.upper().str.strip() to ensure all state codes were uniform, uppercase, and free of leading/trailing whitespace.
- **Missing Value Analysis**: Conducted a systematic check for null entries in geographic columns. Any rows missing a state or incident label were dropped using .dropna() to prevent inaccuracies in the final counts.

4.2. State Name Normalization (Mapping Dictionary)

A critical validation step involved aligning the FEMA dataset with external GeoJSON geometry.

- **The "Grey Map" Fix**: Since the dataset utilizes two-letter ISO codes (e.g., "TX") and the GeoJSON utilizes full names (e.g., "Texas"), a mapping dictionary was implemented.
- **Data Consistency**: This normalization ensured a 1:1 match rate, preventing states from appearing as empty "cold spots" due to naming mismatches.

4.3. Spatial Join & Merge Validation

Once normalized, the data was validated through a trial merge with **GeoPandas**.

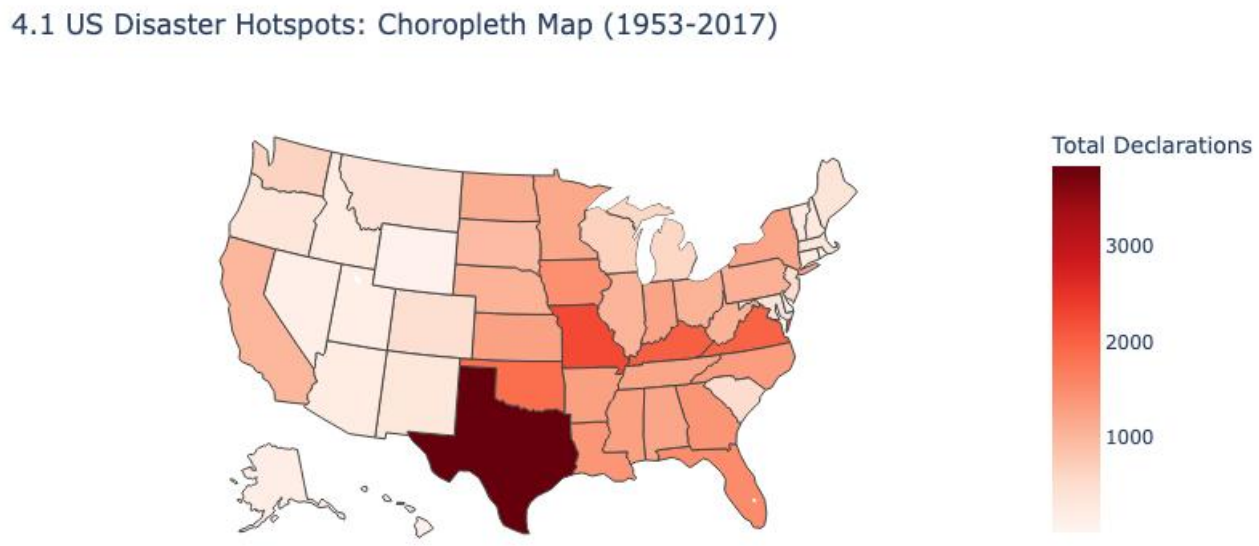
- **Integrity Check**: Confirmed that the result of the merge operation maintained the correct number of state polygons.
- **Handling Zeros**: Applied .fillna(0) to the merged GeoDataFrame, ensuring that states with no recorded incidents for a specific category were still rendered on the map rather than disappearing from the visualization.

5. Temporal Trend Analysis

The primary focus of this phase is to analyze how disaster declarations are distributed across the United States. By identifying spatial patterns and hotspots, we can justify why certain regions are more vulnerable to specific environmental threats.

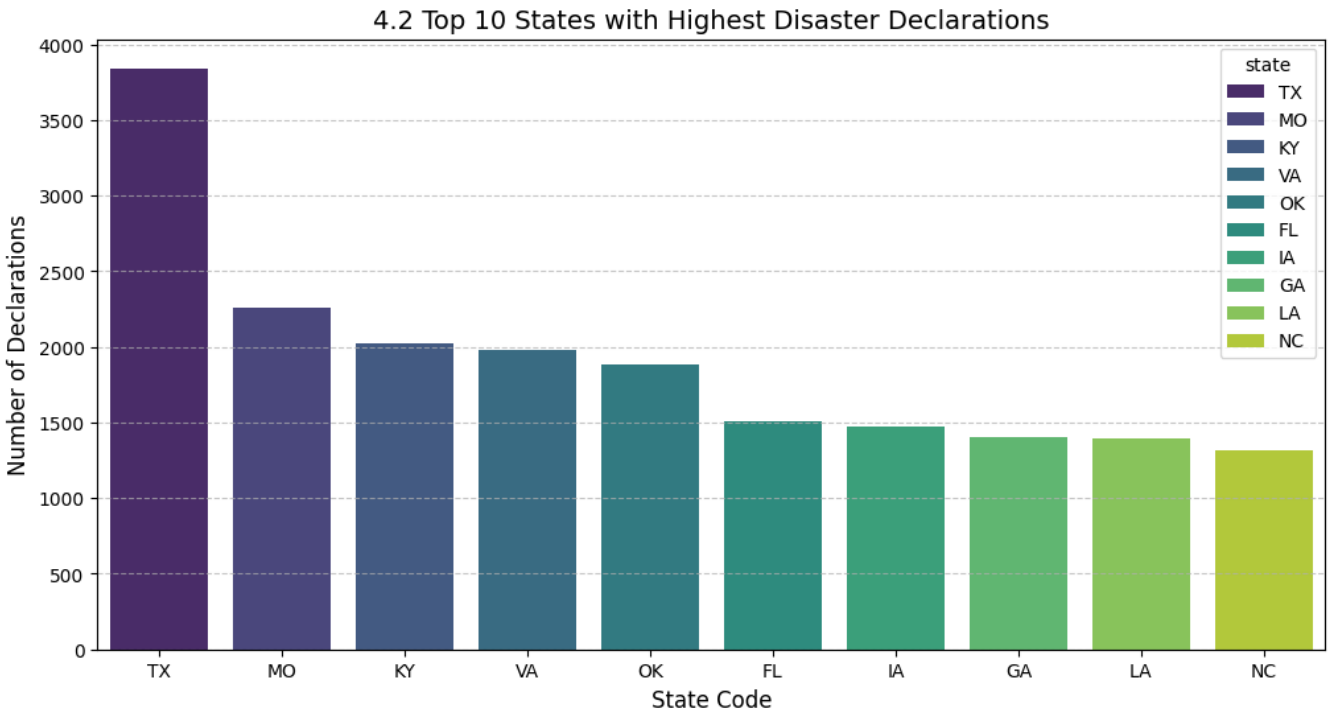
5.1. Nationwide Distribution (Plotly Choropleth)

- **Task:** Aggregate total disasters per state and visualize them using a dynamic, interactive choropleth map.
- **Represents:** The total frequency of all federal disaster declarations across the 50 US states.
- **Spatial Pattern:** A distinct high-intensity cluster is observed in the **Southern and Western United States**, with **Texas** and **California** appearing in the darkest shades.
- **Explanation:** This pattern is driven by large geographic landmasses combined with exposure to diverse risk zones like the Gulf Coast and Pacific wildfire corridors.



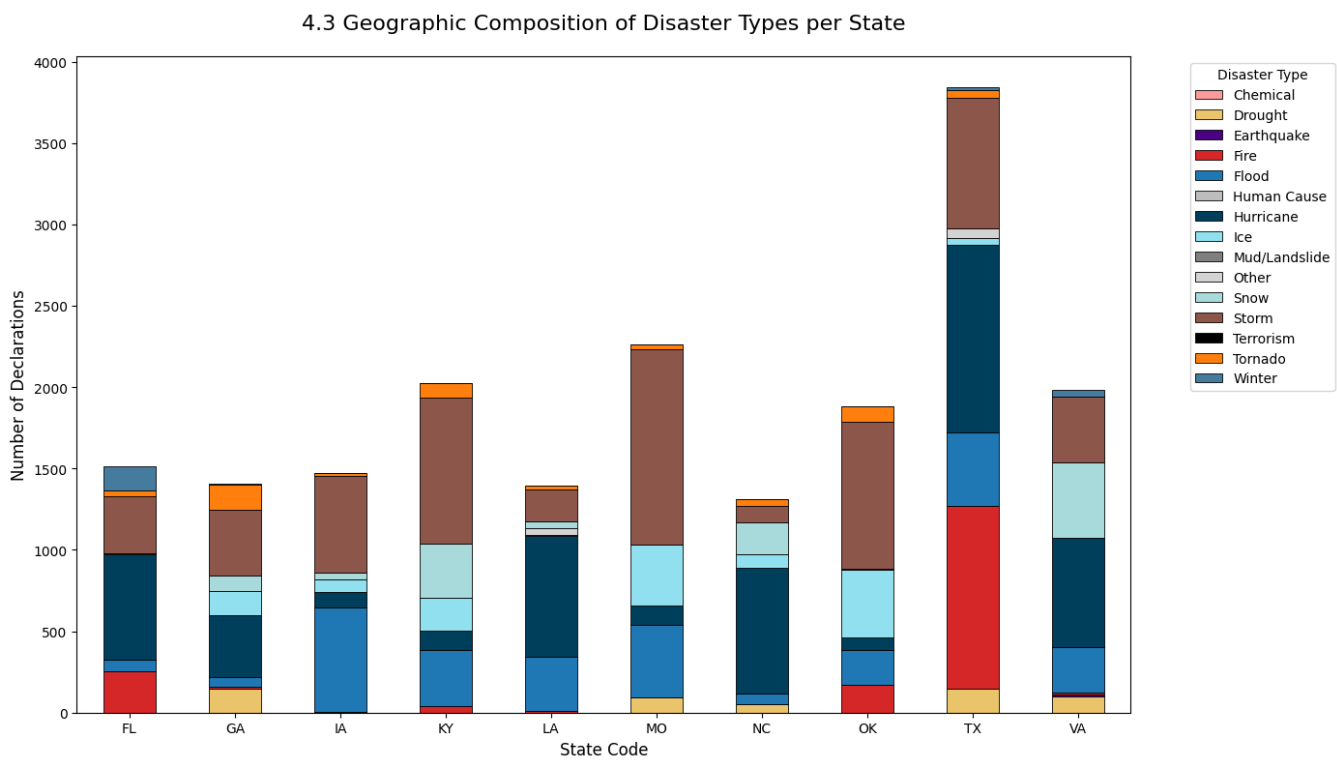
5.2. Top 10 State Comparative Analysis (Seaborn)

- **Task:** Identify the ten states with the highest total disaster counts and visualize them using a focused bar chart.
- **Represents:** A direct comparison of the states carrying the heaviest cumulative disaster burden.
- **Spatial Pattern: Texas** is a clear national outlier, followed by a secondary group (KY, MO, FL) with remarkably similar declaration frequencies.
- **Explanation:** The dominance of Texas is due to its size crossing multiple climate zones, while the similarity between MO and KY reflects shared vulnerability to regional storm and flood cycles.



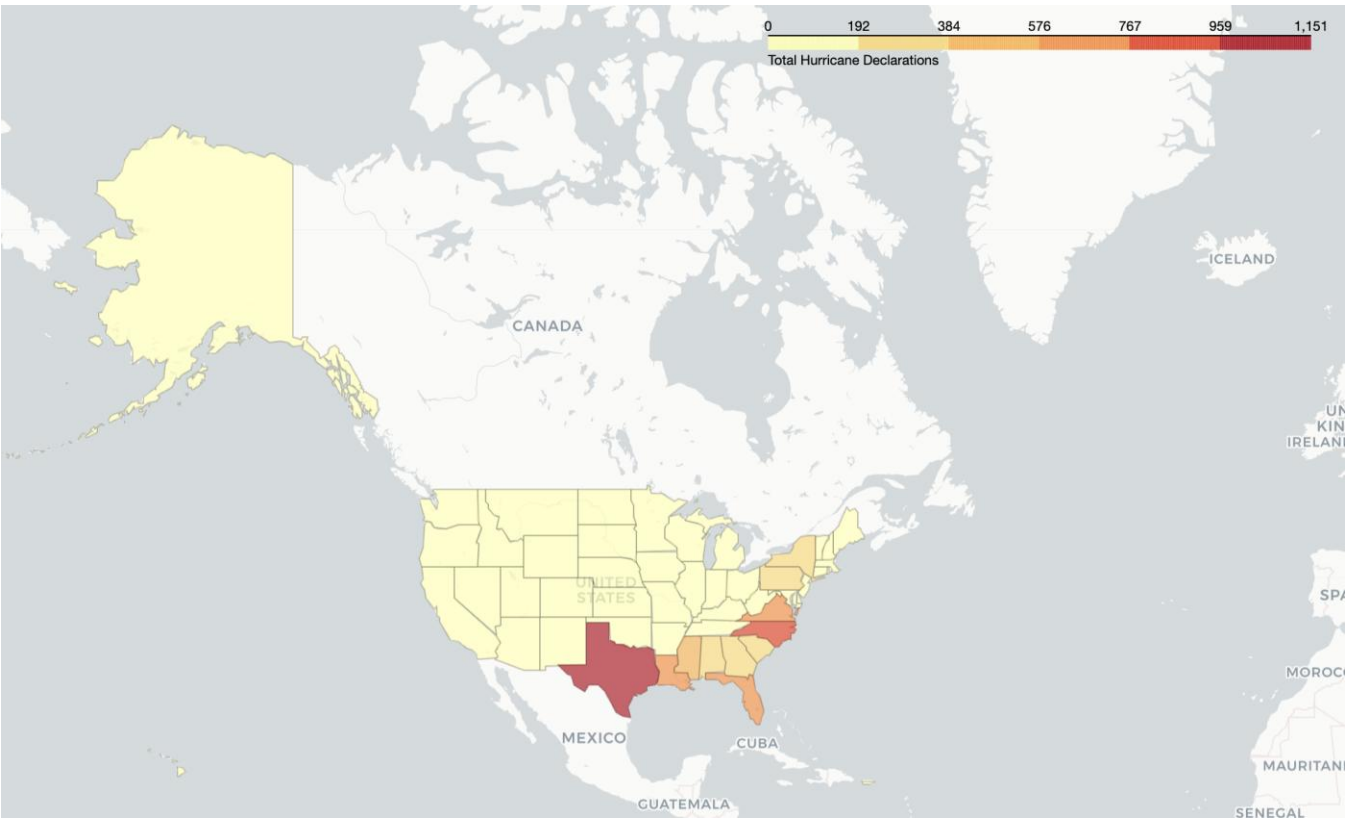
5.3. Geographic Incident Composition (Stacked Bar)

- **Task:** Breakdown the proportional "mix" of disaster types for the top 10 states using a custom color-mapped stacked bar chart.
- **Represents:** The specific disaster "signature" of each state (e.g., what percentage of their declarations are Fire vs. Flood).
- **Spatial Pattern: California** is dominated by **Fire (Red)**, while **Florida** and **North Carolina** are dominated by **Hurricanes (Dark Blue)**.
- **Explanation:** This confirms that disaster risk is region-specific; arid Western climates drive fire events, whereas coastal proximity drives tropical water-based disasters.



5.4. Interactive Hurricane Hotspot Detection (Folium)

- **Task:** Create a specialized geospatial intensity map focusing on a single high-impact disaster type (**Hurricane**) using **Folium** and **GeoPandas**.
- **Represents:** An interactive exploration of where specific disaster types are geographically concentrated.
- **Spatial Pattern:** A massive hotspot is visible along the **Gulf Coast** and the **Southeastern Atlantic Seaboard**.
- **Explanation:** Hurricanes are fueled by warm ocean waters, which naturally restricts their primary impact to coastal landfall zones rather than landlocked states.



6. Visualization Standards

To ensure technical accuracy and readability, every plot generated in this geospatial analysis adheres to a strict set of visualization standards. These standards ensure that complex spatial insights are communicated clearly and professionally.

Mandatory Elements Included:

- **Standardized Legends**: Every map and chart includes a legend to link color intensity or category to real counts, ensuring that visual data is quantifiable.
- **Axis Labeling**: Both X-axis and Y-axis are explicitly labeled with units (e.g., State Code, Number of Declarations).
- **Clutter Reduction**: Bar chart labels are kept horizontal for clarity, and font sizes are scaled to avoid overlapping data points.
- **Interactive Elements**: Tooltips and zoom functions are enabled for **Plotly** and **Folium** maps, allowing for detailed data exploration and precise value identification upon hovering.
- **Color Accuracy**: Custom color mapping (e.g., Red for Fire, Blue for Flood) was implemented in the stacked bar charts to create an intuitive and consistent visual language across the report.

7. Conclusion

The temporal analysis phase of the US Natural Disaster Visualization project was completed successfully. By following the structured workflow in Milestone-2, we successfully validated the historical dataset and extracted meaningful time-based features.

Key Outcomes of this Phase:

- **Regional Risk Mapping**: Successfully identified long-term geographic growth patterns and spatial clusters in disaster declarations.
- **Disaster Signature Identification**: Developed a clear understanding of how different disaster types evolve in specific regions, such as wildfires in the West versus hurricanes on the East Coast.
- **Interactive Hotspot Mapping**: Established a baseline for recurring geographic patterns, providing a foundation for future predictive modeling and resource allocation

The combination of validated geographic data, standard-compliant visualizations, and detailed regional insight ensures that this project provides a reliable overview of disaster risks in the United States.